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STUDENT NUMBER: _____

TEACHER: _____

THE HILLS GRAMMAR SCHOOL

TERM ONE ASSESSMENT TASK 2016

YEAR 11

MATHEMATICS

Time Allowed: 55 minutes

Weighting: 15%

Outcomes: P1, P2, P3, P4

Class Teachers: Mr O'Neill, Mr Parrish, Mrs Singh, Mr Tobin

Instructions:

- Approved calculators may be used
- Attempt all questions
- Start all questions on a new sheet of paper
- The marks for each question are indicated on the examination
- Show all necessary working

Topic Outcome	Multiple Choice	Question 1	Question 2	Question 3	Question 4	TOTAL
Marks	3	19	14	9	10	55

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Section 1 – Multiple Choice (3 marks)

Questions 1 to 3 are multiple choice. Place your answers in the answer grid below.

1) Fully simplify $\frac{x^2 - 4}{x + 2}$

A. $\frac{x - 2}{x + 2}$

B. $\frac{x^2 - 4}{x}$

C. $2x + 4$

D. $2x - 4$

2) Fully simplify $\sqrt{18} - \sqrt{50}$

A. $\sqrt{2}$

B. $3\sqrt{50}$

C. $4\sqrt{3} - 3\sqrt{2}$

D. $-2\sqrt{2}$

3) What is the value of $\sqrt{4.01^2 - 0.8^2}$ correct to four significant figures?

(A) 1.618

(B) 1.691

(C) 1.619

(D) 1.62

MULTIPLE CHOICE ANSWER GRID

1	(A)	(B)	(C)	(D)
2	(A)	(B)	(C)	(D)
3	(A)	(B)	(C)	(D)

Section 2 – Extended Response (52 marks)*Answer questions 1 – 4 on the lined paper provided.*

Question 1 (19 Marks)	Marks
a) Simplify the following:	
i $-2x^2 + 3x - 4x^2 - 5x$	2
ii $(-3x^2)^3$	2
b) Expand and simplify the following:	
i $8 - 4(2y + 1) + y$	2
ii $(x - \frac{1}{x})(x + \frac{1}{x})$	2
c) Fully factorise the following:	
i $2y^2 - 11y - 6$	2
ii $27a^3 - 64$	3
d) Simplify the following algebraic fractions:	
i $\frac{x^3 + 3x^2 - 9x - 27}{x^2 + 6x + 9}$	3
ii $\frac{2}{x^2 - 4} - \frac{3}{x + 2}$	3

Question 2 (14 Marks)**Marks******start this question on a new page****

a) Solve the following equations:

$$\text{i} \quad \frac{x+6}{4} = \frac{2x-3}{3}$$

3

$$\text{ii} \quad x^2 - 15x = 16$$

2b) Solve using the **quadratic formula**: $3x^2 - 5x + 1 = 0$.**3**

Leave your answer in simplified surd form.

c) Solving by **completing the square**: $x^2 - 2x = 1$.**3**

Leave your answer in simplified surd form.

d) Solve the following pair of simultaneous equations.

3

$$xy = 6$$

$$x + y = 5$$

Question 3 (9 Marks)**Marks******start this question on a new page****

a) Simplify the following:

$$\text{i.} \quad \frac{2\sqrt{3}}{7\sqrt{6} - \sqrt{54}}$$

3

$$\text{ii.} \quad (\sqrt{3} + 2\sqrt{3})(\sqrt{3} - 2\sqrt{3})$$

2b) Find the exact value of a and b if $\frac{\sqrt{3}-4}{2+\sqrt{3}} = a + b\sqrt{3}$ **3**

Question 4 (10 Marks)**Marks******start this question on a new page****

a) Find the natural domain and range of each of the following functions.

i.	$h(x) = x^3$	2
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ii.	$y = \frac{3}{x-2}$	2
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b) Given $f(x) = \begin{cases} x^2 - 3x & \text{if } x > 2 \\ x & \text{if } x \leq 2 \end{cases}$

find

i)	$f(6) - f(-2)$	2
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ii)	Sketch the graph of $f(x)$	3
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iii)	Is $f(x)$ a function or a non-function?	
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Give a reason.	1
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END OF ASSESSMENT TASK

Year 11 MATHEMATICS TASK ONE 2016

THGS Exam Feedback

Suggested Solutions, Marking Scheme and Markers' comments

<u>Suggested solution(s)</u>	<u>comments</u>
1. $\frac{2(x-2)(x+2)}{(x+2)} = 2x-4$ (D) 2. $\sqrt{18} - \sqrt{50}$ $= 3\sqrt{2} - 5\sqrt{2}$ $= -2\sqrt{2}$ (D) 3. 1.619 (C)	Generally very well answered.
<u>Section 2</u> <u>Question 1</u> a) (i) $(-2x^2 + 3x - 4x^2 - 5x)$ $= -6x^2 - 2x$ (1) (1) (ii) $(-3x^2)^3 = -27x^6$ (1) (1) b) (i) $8 - 4(2y+1) - y$ $= 8 - 8y - 4 - y$ (1) $= 4 - 9y$ (1)	Some students factorised but this was not necessary. common error was -9 instead of -27 generally well answered

Suggested Solutions, Marking Scheme and Markers' comments

Suggested solution(s)	comments
$(ii) \left(x - \frac{1}{x}\right) \left(x + \frac{1}{x}\right)$ $= x^2 - \frac{1}{x^2} \quad (1)$	well answered
$c) (i) 2y^2 - 11y - 6$ $= 2y^2 - 12y + 1y - 6$ $= 2y(y-6) + 1(y-6) \quad (1)$ $= (2y+1)(y-6) \quad (1)$	well answered
$(ii) 27a^3 - 64$ $= (3a)^3 - 4^3 \quad (1)$ $= (3a-4)(9a^2+12a+16) \quad (1)$	Some substitute substituted $27a$ instead of $3a$. Also '+' and '-' signs in wrong place
$d) (i) \frac{x^3 + 3x^2 - 9x - 27}{x^2 + 6x + 9}$ $= \frac{x^2(x+3) - 9(x+3)}{(x+3)^2} \quad (1)$ $= \frac{(x^2-9)(x+3)}{(x+3)(x+3)} \quad (1)$ $= \frac{(x-3)(x+3)(x+3)}{(x+3)(x+3)} = x-3 \text{ simplify.} \quad (1)$	Most common error was students did not factorise x^2-9 therefore did not fully

Suggested Solutions, Marking Scheme and Markers' comments

Suggested solution(s)	comments
(ii) $\frac{2}{(x-2)(x+2)} - \frac{3}{x+2}$ ① $= \frac{2 - 3(x-2)}{(x-2)(x+2)}$ $= \frac{2 - 3x + 6}{(x-2)(x+2)}$ ① $= \frac{8 - 3x}{(x-2)(x+2)}$ ①	Generally well answered. Some had this as '-' and therefore had $-4 - 3x$.
<u>Question 2</u> a) (i) $\frac{x+6}{4} = \frac{2x-3}{3}$ $3(x+6) = 4(2x-3)$ ① $3x+18 = 8x-12$ ① $-5x = -30$ ① $x = 6$ ①	Generally well answered. Cross multiplying was the best utilised technique.

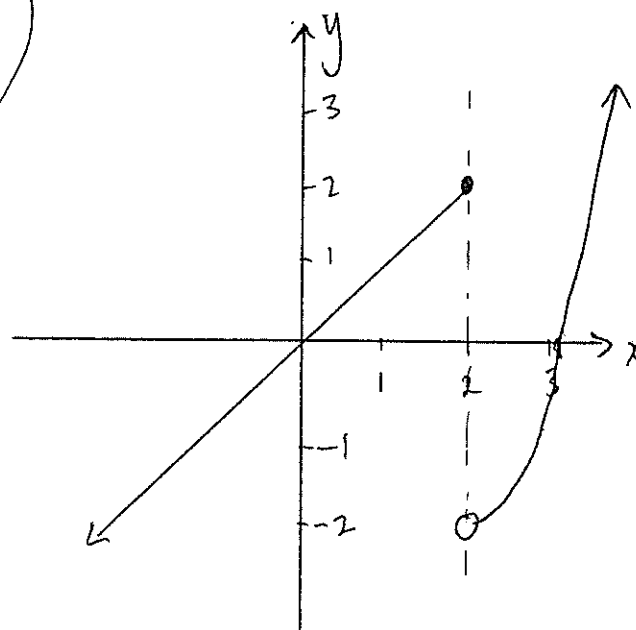
Suggested Solutions, Marking Scheme and Markers' comments

Suggested solution(s)	comments
(ii) $x^2 - 15x = 16$ $x^2 - 15x - 16 = 0$ $(x - 16)(x + 1) = 0 \quad (1)$ $x = 16, -1 \quad (1)$	Well answered
b) $3x^2 - 5x + 1 = 0$ $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ $= \frac{5 \pm \sqrt{25 - 4 \times 3 \times 1}}{2 \times 3} \quad (1)$ $= \frac{5 \pm \sqrt{13}}{6} \quad (2)$	Some students failed to use the '+' sign in front of the $\sqrt{\quad}$ (-1 mark).
c) $x^2 - 2x = 1$ $x^2 - 2x + (-1)^2 = 1 + (-1)^2 \quad (1)$ $(x - 1)^2 = 2$ $x - 1 = \pm \sqrt{2} \quad (1)$ $x = \pm \sqrt{2} + 1 \quad (1)$	Most students successfully used the completing the square method to solve this.

Suggested Solutions, Marking Scheme and Markers' comments

Suggested solution(s)	comments
1) $xy = 6$ (1) $x + y = 5$ (2) $y = 5 - x$ (3) $5x - x^2 = 6$ (1) $x^2 - 5x + 6 = 0$ $(x-3)(x-2) = 0$ $x = 3, 2$ (1) $y = 2, 3$ (1) $\therefore (2, 3) (3, 2)$	The main problem in this question was students not stating which <u>x values corresponded to the y values.</u> Two correct ways: ① $(2, 3)$ and $(3, 2)$ ② when $x=2, y=3$ when $x=3, y=2$
<u>Question 3</u> a)(i) $\frac{2\sqrt{3}}{7\sqrt{6} - \sqrt{54}} = \frac{2\sqrt{3}}{7\sqrt{6} - 3\sqrt{6}} = \frac{2\sqrt{3}}{4\sqrt{6}}$ (1) (1) $= \frac{1}{2\sqrt{2}} = \frac{\sqrt{2}}{4}$ (1) $\rightarrow$ must rationalise denom to get 3rd mark. (ii) $(\sqrt{3} + 2\sqrt{3})(\sqrt{3} - 2\sqrt{3})$ $= 3 - 12 = -9$ (1)	<u>Q3</u> • question generally poorly ans by 2u cohort. • ans well.

Suggested Solutions, Marking Scheme and Markers' comments

Suggested solution(s)	comments
b) $\frac{\sqrt{3}-4}{2+\sqrt{3}} \times \frac{2-\sqrt{3}}{2-\sqrt{3}} = \frac{2\sqrt{3}-8-3+4\sqrt{3}}{4-3}$ $= 6\sqrt{3}-11$ ① $\therefore a = -11, b = 6$ ①	• generally, poorly answered. Most did not know what to do with a and b
<u>Question 4</u>	Must state
a) (i) domain: all real x or $x \in \mathbb{R}$ ① range: all real y or $y \in \mathbb{R}$ ①	$a = b =$ Students made a large number of mistakes in this question. Statements need to be made
(ii) domain: $x \neq 2$ ① range: $y \neq 0$ ①	eg domain $x \in \mathbb{R}$
b) (i) $f(6) - f(-2) = 36 - 18 - (-2)$ $= 20$ ①	Students did not realise that this was a composite function
(ii)  ①-shape ①-circle ①-value	Sketch should be $\frac{1}{2}$ page
(iii) Yes, satisfies vertical line test	Students did not refer to vertical line test